

Week 5 Project Summary

Programme: AnalystLab Africa Experience Lab

Project: HealthConnect Clinic Experience Lab (Improving Patient Appointment Attendance and Healthcare Support Using Data and AI)

Track: Data Analytics

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Week: Week 5, Solution Development and Implementation

1. Week 4's expectations for Week 5

Week 4 left three jobs for Week 5:

1. Calculate and chart the five KPIs (Week 4 only defined them).
2. Check whether the three strongest variables still hold up when looked at together.
3. Agree the outcome-variable definition with the Data Science track before they model.

2. The Week 5 deliverables

- Prepared the data on a copy: checked types, missing values, duplicates, category consistency, and cross-field logic. The original CSV was never touched.
- Ran the EDA across all eight dimensions the brief asks for: appointment characteristics, patient characteristics, previous history, reminders, waiting time, distance, cancellations, and no-show outcomes.
- Ran a confounding check. The three strongest variables turned out to be independent risk factors, not one signal showing up three times.
- Calculated, charted, and interpreted all five KPIs.
- Built a six-panel dashboard covering the headline number and the five KPIs, first in Python.
- Wrote up five business insights with what each one means for the clinic, five recommendations, and a limitations section.
- Identified and prepared the Data Analytics to Data Science handover (see `CrossTrack_Collaboration.md`).
- Rebuilt the same dashboard in Power BI as a second version, using a sort-order-corrected export of the processed data.

3. Key findings

- The overall no-show rate is 48.5%. Nearly half of every booked slot goes unused, against just 5.3% that are properly cancelled.
- The two strongest predictors are booking lead time (27.8% no-show rate at 0-7 days, rising to 67.7% at 46-60 days) and prior no-show history (43.5% at zero prior misses, rising to 68.8% at three or more). They also stack: a patient with two or more prior no-shows and a 31+ day lead time misses 73.0% of the time. That group is only 4.7% of all bookings.
- Distance only bites past a point. The rate is flat up to 15 km, then jumps to 54.1% beyond 15 km.

- Reminders help a little, about 4 percentage points, but they are not the fix on their own.
- The cancellation-to-no-show ratio drops as lead time grows (0.202 down to 0.076). Patients who book far ahead are less likely to warn the clinic, which is why Cancelled and No-Show should stay separate.

4. Challenges

- Getting a reproducible Python setup running on version 3.14, which is new enough that I had to check each package would install before settling on the toolchain.
- Choosing how finely to band the continuous variables. I sized every band by row count so no KPI leans on a group that is too small to trust, and where one group was unavoidably small (the "3+" prior no-show bracket, 93 rows) I flagged it in the notebook rather than hiding it.
- Power BI sorted the band columns alphabetically by default, which would have shown two of the three KPI charts in the wrong order. I added numeric sort-order columns to the export to fix it.

5. Decisions and why

- Rows with a missing distance are excluded from distance KPIs, not filled in. Filling them would add made-up values into a variable that actually predicts no-shows. This carries over from the Week 4 plan and still holds.
- `appointment_outcome` stays as three classes. Week 4 argued this on principle; Week 5 has the KPI 5 evidence to back it.
- I built the dashboard in Python first because the analysis was already there, then rebuilt it in Power BI rather than picking only one tool.

6. Changes to the Week 4 approach

The KPI definitions, the business questions, and the missing-data handling are all unchanged. Week 5 ran the Week 4 plan as written. The one addition is the Power BI dashboard. Week 4 had pushed visualisation tooling to a later phase, and I brought it forward so there is a more presentation-ready version.

7. Cross-track collaboration

The Week 5 brief asks each track to identify a dependency with another track and says this is meant to simulate a multidisciplinary environment. The dependency here is Data Analytics to Data Science: they will model on the same dataset, so their outcome-variable choice depends on the categories being checked first. I prepared that handover: the three-class outcome recommendation with KPI 5 as evidence, the three validated independent features, and the processed dataset with documented band boundaries. Working out what they would need also shaped my own analysis (I built KPI 5 and ran the confounding check partly for this). Full write-up in `CrossTrack_Collaboration.md`.

8. Remaining work

- The predictive model is to be built by the Data Science track.
- The distance and prior-no-show thresholds should be rechecked once there is more data, since the "3+" bracket and the 15+ km band are based on smaller samples.

9. Proposed focus for Week 6

- Track the five KPIs against fresh data as it arrives and see whether these patterns hold over time.
- Once the Data Science track has a baseline model, check whether the three validated features actually carry weight in it, not just in a rate comparison.
- Look harder at the double-risk group (two or more prior no-shows plus a 31+ day lead time). It is small at 4.7% of bookings but has the highest no-show rate by a wide margin at 73.0%, so it may be worth designing a specific intervention for later.